Root Cause Analysis (RCA) Report - AIOps Lab 4

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1. Executive Summary

This report outlines the Root Cause Analysis (RCA) of an automated incident detected by the AIOps Machine Learning pipeline. The anomaly was initially flagged by the Isolation Forest anomaly detection model (Lab 3) operating on real-time system telemetry. Subsequent automated analysis successfully isolated the underlying cause to a specific endpoint overload, leading to severe latency degradation and timeout failures.

2. Incident Overview & Structured RCA Data

The automated RCA engine analyzed the generated logs and metrics during the anomaly window, correlating latency, request rates, and error categories. The findings are documented in the structured output below:

- **Incident ID:** INC-RCA-362B6483
- **Timestamp of Analysis:** 2026-04-15
- **Root Cause Endpoint:** /api/slow
- **Primary Trigger Signal:** Latency
- **Confidence Score:** 95%

Supporting Evidence (Signal Analysis)

During the anomaly window, the following critical metrics were recorded:

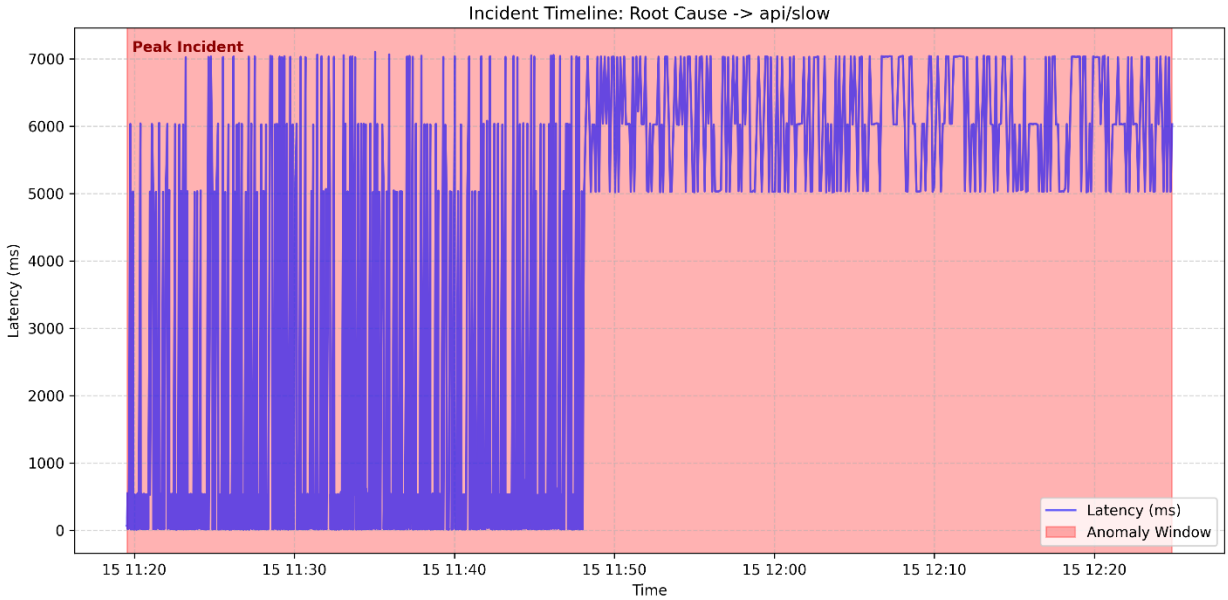
- **Peak Latency Recorded:** 7,103.11 ms
- **Average Latency During Window:** 3,272.15 ms
- **Total Errors in Window:** 358 errors
- **Predominant Error Category:** TIMEOUT_ERROR

3. Endpoint Attribution & Error Category Analysis

Based on the signal analysis, the endpoint /api/slow was definitively identified as the primary contributor to the anomaly. The surge in requests to this endpoint exhausted system resources, causing a bottleneck.

This bottleneck manifested primarily as a massive spike in TIMEOUT_ERROR logs. The distribution of error categories confirms that the system did not suffer a database crash or a code syntax error, but rather an infrastructure capacity failure induced by prolonged execution times on the isolated endpoint.

4. Incident Timeline Visualization



Timeline Narrative

As illustrated in the timeline visualization above, the incident followed a clear trajectory:

1. **Normal State:** The system was operating nominally with latency consistently resting below the 1000ms threshold.
2. **Anomaly Start:** A sudden influx of traffic targeted at the /api/slow endpoint caused the initial latency spike, breaching the dynamic baseline set by the Machine Learning model. The window is highlighted in red.

3. **Peak Incident:** The system reached critical degradation, with latency peaking at approximately 7,103 ms. At this stage, the maximum volume of TIMEOUT_ERROR exceptions was logged.
4. **Recovery:** Following the cessation of the targeted traffic surge (or the intervention of the automated remediation engine), the system flushed the pending queue, and latency stabilized back to the normal baseline.

5. Recommended Actions & Remediation

To prevent the recurrence of this specific anomaly, the automated engine recommends the following operational actions:

1. **Immediate Remediation:** Apply strict Rate Limiting policies specifically targeting the /api/slow endpoint to prevent resource exhaustion during unexpected traffic spikes.
2. **Long-term Optimization:** Investigate the backend logic governing the /api/slow route. The persistent TIMEOUT_ERROR signals indicate that the execution logic is synchronous and blocking. Implementing asynchronous processing or aggressive caching for this route is highly advised.